Week 03
Monte Carlo Methods

STUDY-RL — a self-paced Deep RL curriculum

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Learning objectives

Study companion. Read `teaching_guide.md` in this folder for step-by-step examples, an analogy with its limits, and self-checks. Then use `workbook.ipynb` to implement and vary the calculations.

Key Idea

By the end of this week you will be able to:

1. Distinguish *first-visit* and *every-visit* Monte Carlo prediction; explain first-visit unbiasedness, every-visit finite-sample bias and consistency, and why neither has universal variance dominance.
2. Implement *on-policy MC control with an ϵ-soft policy* and reproduce Sutton & Barto's figure 5.2 on Blackjack.
3. Derive the *ordinary* and *weighted* importance-sampling estimators for off-policy MC and compute their variance.
4. Identify when an off-policy estimator is consistent and what *coverage* assumption is required for the importance weights to be defined.
5. Apply off-policy MC to a delayed-feedback *recruitment funnel* as a toy applied matching task.

1 Motivation & intuition

The big idea this week is replacing the *expectation* in the Bellman equation by a *sample mean*. Last week we had the model and could exhaustively compute $\sum_{s',r} p(s', r \mid s, a)[r + \gamma V(s')]$. Now suppose the only thing we have is the ability to *generate* trajectories: sequence of (S_t, A_t, R_{t+1}) tuples by interacting with the environment. The law of large numbers tells us that an average of returns $G^{(1)}, \ldots, G^{(N)}$ obtained from N independent episodes converges to $V^\pi(s_0)$. That is Monte Carlo policy evaluation in one sentence.

Intuition. Monte Carlo methods trade *model knowledge* for *samples*. They are the dynamical-systems equivalent of estimating π by throwing darts: noisy, embarrassingly parallel, and convergent at rate $1/\sqrt{N}$ rather than γ^k. The variance penalty is real; the modelling freedom is enormous.

Why MC and not TD? TD methods (Week 4) combine the Bellman recursion with sampling and so reduce variance *at the cost of bias* (the TD target uses an old, biased value estimate). MC keeps the full episodic return; it is unbiased but high variance. For tasks with short episodes and sparse terminal rewards (e.g. Blackjack, or the recruitment funnel later in this lecture) MC is often the cleaner choice.

Common pitfall

MC and continuing tasks. For a continuing task, a full episode may never be available. With $\gamma = 1$, existence of an integrable return requires additional conditions; almost-sure termination alone does not ensure finite expectation. Truncating a discounted return introduces tail error. Choosing a new discount changes the objective rather than biasing an estimator of that new objective. Average-reward MC exists and is discussed in Mahadevan (1996) and Bertsekas–Tsitsiklis (1996, Ch. 7).

2 Mathematical formulation

2.1 MC policy evaluation

Definition 2.1.1 (First-visit MC estimator)**.** Given N independent episodes generated by π, let $G_s^{(i)}$ be the return following the *first* visit to state s in episode i (skip episodes that never visit s). Then

$$\hat{V}_{\mathrm{FV}}(s) = \frac{1}{N_s} \sum_{i:\, s \in \tau_i} G_s^{(i)},$$

where N_s is the number of episodes in which s was visited at least once.

Definition 2.1.2 (Every-visit MC estimator)**.** $\hat{V}_{\mathrm{EV}}(s) = \frac{1}{M_s} \sum_{(\text{visits to } s)} G_{\text{at that visit}}$, where M_s is the total number of visits, summed over episodes.

Theorem 2.1.3. *$\hat{V}_{\mathrm{FV}}$, conditional on at least one contributing episode, is an unbiased estimator of $V^\pi(s)$ for independent episodes under a fixed stationary Markov policy and integrable returns; $\hat{V}_{\mathrm{EV}}$ is biased in finite samples but consistent.*

Sketch. For first-visit: the random variable $G_s^{(i)}$ has expectation $V^\pi(s)$ by definition of V^π, and the $G_s^{(i)}$ across episodes are i.i.d., so the sample mean is unbiased. For every-visit, repeated visits within the same episode can share environment randomness. Correlation alone does not prove bias; the random visit-count denominator is also coupled to the returns. Under suitable episodic moment assumptions, a law of large numbers across independent episodes gives consistency of the ratio of total returns to total visits. Within-episode dependence need not vanish. See Singh & Sutton 1996. □

2.2 On-policy MC control with ϵ-soft policies

We follow Sutton & Barto's GPI template: alternate (i) MC policy evaluation of Q^π and (ii) policy improvement to an ϵ-greedy update of π:

$$\pi_{\mathrm{new}}(a \mid s) = \begin{cases} 1 - \epsilon + \frac{\epsilon}{|\mathcal{A}|} & \text{if } a = \arg\max_a Q(s,a), \\ \frac{\epsilon}{|\mathcal{A}|} & \text{otherwise.} \end{cases}$$

A positive ϵ gives each action positive probability at visited states. Coverage of state-action pairs additionally requires reachability from the start distribution and sufficiently many visits; unreachable states are not made reachable by action randomisation.

Theorem 2.2.1 (Policy improvement for ϵ-soft policies)**.** *Let π be ϵ-soft and let π' be the ϵ-soft greedification of Q^π above. Then $V^{\pi'}(s) \geq V^\pi(s)$ for all s.*

Idea. Same telescoping argument as Week 2 with the modification that the maximum is taken *within* the class of ϵ-soft policies; the $\epsilon/|\mathcal{A}|$ slack in each action just adds a constant offset that cancels in the telescope. □

2.3 Off-policy MC: importance sampling

Now the data are generated by a *behaviour* policy b, but we want to estimate the value of a different *target* policy π. The *coverage* assumption $b(a \mid s) > 0$ whenever $\pi(a \mid s) > 0$ is required; otherwise the importance weights are undefined.

Definition 2.3.1 (Importance-weight for a trajectory τ)**.**

$$\rho_\tau \triangleq \prod_{t=0}^{T-1} \frac{\pi(A_t \mid S_t)}{b(A_t \mid S_t)}.$$

Theorem 2.3.2 (Ordinary IS)**.**

$$\hat{V}_{\mathrm{OIS}}(\pi) = \frac{1}{N} \sum_{i=1}^{N} \rho_{\tau_i} G^{(i)}$$

is an unbiased *estimator of* $V^\pi(s_0)$ *under coverage, but its variance can be infinite.*

Sketch. $\mathbb{E}_b[\rho_\tau f(\tau)] = \sum_\tau b(\tau) \frac{\pi(\tau)}{b(\tau)} f(\tau) = \mathbb{E}_\pi[f(\tau)]$ for any bounded f, by direct change of measure. Set $f(\tau) = G(\tau)$. Variance: $\mathrm{Var}_b[\rho_\tau G] = \mathbb{E}_\pi[\rho_\tau G^2] - V^\pi(s_0)^2$ where ρ_τ inside the π-expectation is now upper-bounded by $(\max_{s,a} \pi/b)^T$; for T large this can blow up exponentially. □

Theorem 2.3.3 (Weighted IS)**.**

$$\hat{V}_{\mathrm{WIS}}(\pi) = \frac{\sum_i \rho_{\tau_i} G^{(i)}}{\sum_i \rho_{\tau_i}}$$

is biased *(*$\mathbb{E}[\hat{V}_{\mathrm{WIS}}] \neq V^\pi$ *in finite samples) but consistent under coverage and the required integrability assumptions. It is defined only when* $\sum_i \rho_{\tau_i} > 0$*. With nonnegative weights and returns bounded in* $[L, U]$*, the estimate also lies in* $[L, U]$*.*

Remark (Practical guidance). WIS often reduces sensitivity to a fluctuating total weight, but neither variance nor mean-squared-error dominance over OIS is universal. Compare estimator uncertainty and effective sample size, and check coverage. The companion teaching guide gives an exact two-action counterexample.

Per-decision IS. Trajectory-level ρ_τ is exponentially bad in T. *Per-decision* IS noticing that the reward at time t depends only on $A_0, \ldots, A_t$:

$$\hat{V}_{\mathrm{PD}}(\pi) = \frac{1}{N} \sum_i \sum_{t=0}^{T-1} \gamma^t \rho_{0:t}^{(i)} R_{t+1}^{(i)}, \quad \rho_{0:t} = \prod_{k=0}^{t} \frac{\pi(A_k \mid S_k)}{b(A_k \mid S_k)}.$$

Per-decision IS has the same expectation and (usually) much lower variance. See Precup, Sutton, Singh (2000) for the full proof.

3 Algorithms

Algorithm 1: First-visit MC policy evaluation.

Input: π, γ, N episodes.
Initialise $\hat{V}(s) \leftarrow 0$, $N_s \leftarrow 0$ for all s.;
for $i = 1, \dots, N$ **do**
  Generate an episode $S_0, A_0, R_1, \dots, S_T$ following π.;
  $G \leftarrow 0$;
  for $t = T-1, T-2, \dots, 0$ **do**
    $G \leftarrow R_{t+1} + \gamma G$;
    if $S_t \notin \{S_0, \dots, S_{t-1}\}$ *(first-visit)* **then**
      N_{S_t} += 1; $\hat{V}(S_t)$ += $(G - \hat{V}(S_t))/N_{S_t}$;
    end
  end
end

Algorithm 2: On-policy every-visit MC control with ϵ-soft policies.

Input: γ, ϵ, N episodes.
Initialise $Q(s,a) \leftarrow 0$, $N_{s,a} \leftarrow 0$; arbitrary π.;
for $i = 1, \dots, N$ **do**
  Generate an episode following the ϵ-soft π.;
  $G \leftarrow 0$;
  for $t = T-1, \dots, 0$ **do**
    $G \leftarrow R_{t+1} + \gamma G$; N_{S_t,A_t} += 1; $Q(S_t,A_t)$ += $(G - Q(S_t,A_t))/N_{S_t,A_t}$;
  end
  Update π to be ϵ-soft greedification of Q;
end

Algorithm 3: Off-policy MC via importance sampling.

Input: behaviour b, target π, γ, N episodes.
for $i = 1, \dots, N$ **do**
  Generate episode τ_i under b; compute $G^{(i)}$ and $\rho_{\tau_i} = \prod_t \pi(A_t \mid S_t)/b(A_t \mid S_t)$.;
end
return $\hat{V}_{\text{OIS}} = \frac{1}{N}\sum_i \rho_{\tau_i} G^{(i)}$;
return $\hat{V}_{\text{WIS}} = \sum_i \rho_{\tau_i} G^{(i)} / \sum_i \rho_{\tau_i}$;

Common pitfall

Forgetting coverage. If $b(a \mid s) = 0$ on a state-action pair where $\pi(a \mid s) > 0$, the importance weight is $\frac{\pi}{0}$ — the estimator silently returns NaN or ∞. Check the support of the full target and logging policies before doing IS; positive propensities in observed rows alone cannot reveal unsupported actions that never appear in the data. The bug is easiest to introduce when the behaviour policy is a deterministic logged policy.

4 Implementation walkthrough

The Blackjack environment is faithful to Sutton & Barto Example 5.1.

```
# project/blackjack.py (excerpt)
@dataclass
class BlackjackEnv:
    seed: int = 0
    def reset(self, seed=None):
        if seed is not None: self.rng = np.random.default_rng(seed)
        self.player = [_draw_card(self.rng), _draw_card(self.rng)]
        self.dealer = [_draw_card(self.rng), _draw_card(self.rng)]
        while _hand_value(self.player)[0] < 12:
            self.player.append(_draw_card(self.rng))
        self.done = False
        return self._obs()
```

The workbook helper below handles one-step rows (the bandit special case). For a trajectory, construct the product of action ratios explicitly before applying the trajectory-level formula. Its numerical floors avoid division errors but do not validate coverage; check support and a positive total weight before interpreting a result.

```
# project/solutions.py (excerpt)
def exercise_5_weighted_is(rewards, target_prob, logged_prob):
    r  = np.asarray(rewards, dtype=float)
    w  = np.asarray(target_prob) / np.maximum(np.asarray(logged_prob), 1e-12)
    return float((w * r).sum() / max(w.sum(), 1e-12))
```

5 Empirical results

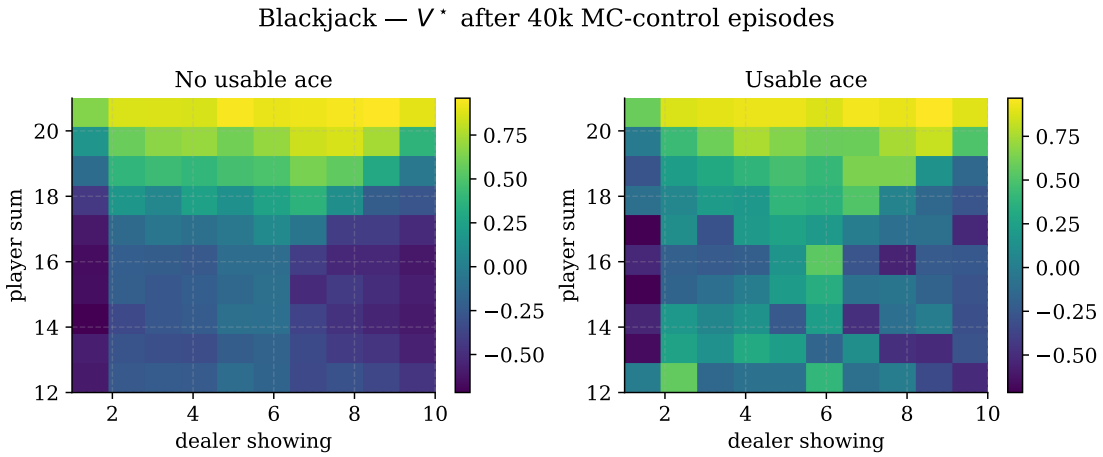


Figure 5.1: Estimated optimal value function on Blackjack after 40k MC-control episodes (with $\epsilon = 0.1$). The left panel is for hands without a usable ace, the right panel for hands with one. The shapes match Sutton & Barto's figure 5.2 (1998).

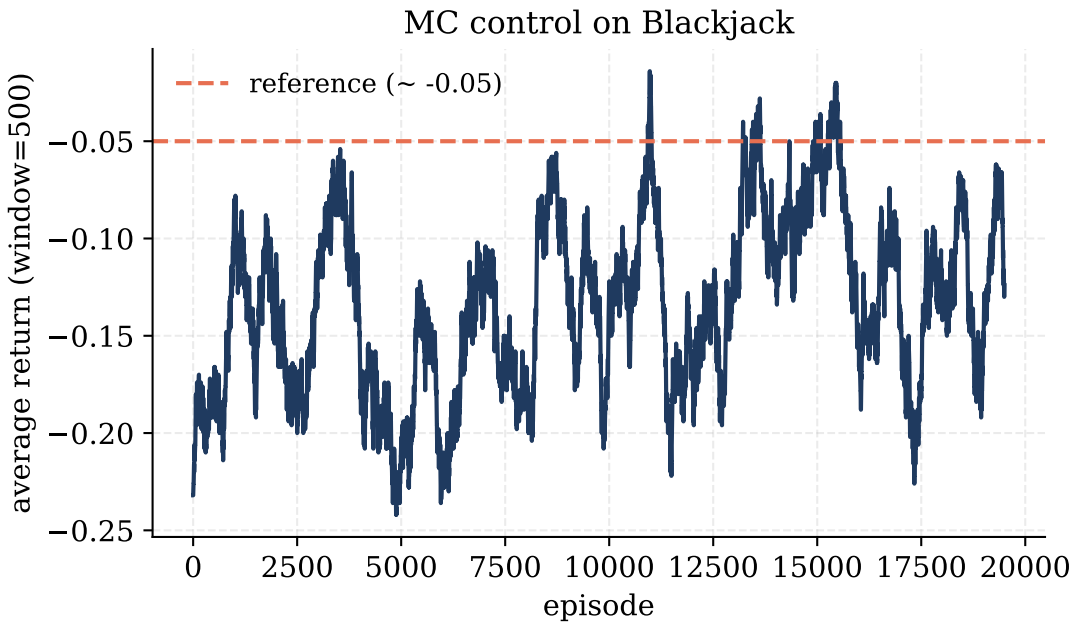


Figure 5.2: 500-episode rolling average of MC-control return on Blackjack.

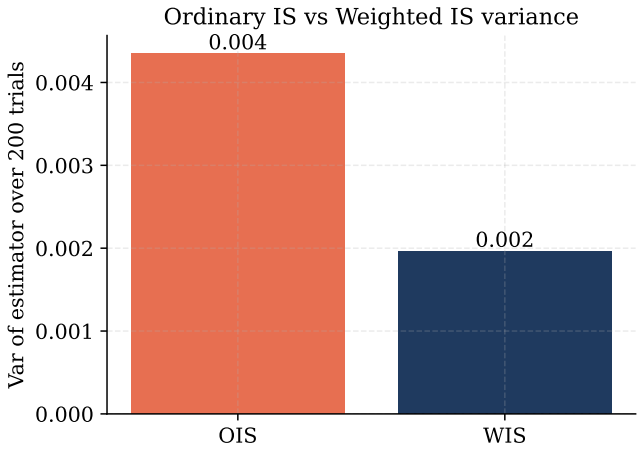


Figure 5.3: Variance of OIS vs WIS over 200 independent trials of a tiny two-arm Bernoulli bandit with $\pi = [0.9, 0.1]$, $b = [0.5, 0.5]$. WIS is many times lower-variance even with weights bounded by 1.8.

6 Safety angle

Safety angle

Off-policy evaluation is one source of evidence for a policy change. It does not alone certify safety. Three failure modes:

1. *Coverage holes.* If the production behaviour policy never routed certain lead-types to certain brokers, the OIS / WIS / DR cannot identify the reward of an unsupported action without additional modelling assumptions. This is a missing-information problem, not merely infinite variance. Mitigation: *forced exploration budget* (a small ϵ of uniform routing) baked into the production policy so logs cover the action space.
2. *Tail importance weights.* A single huge ρ_τ dominates the OIS estimate. Clip the weights, or use WIS, or use the SWITCH / SNIPS estimators (Swaminathan & Joachims 2015). Measure the resulting bias and variance in the chosen setting; no universal mean-squared-error improvement follows.
3. *Reward shaping under delay.* Recruitment outcomes arrive weeks later; if your training pipeline drops not-yet-realised rewards as zero, you bias the evaluator. Use censored-likelihood models (Chapelle 2014; `deeprl_core_sota.md` §12).

Sources. Thomas & Brunskill (ICML 2016), https://proceedings.mlr.press/v48/thomasa16.html, develops weighted doubly robust evaluation and MAGIC. High-confidence off-policy evaluation is a separate problem; see Thomas, Theocharous & Ghavamzadeh (AAAI 2015), https://arxiv.org/abs/1404.3410. A confidence bound needs its own assumptions, including boundedness or tail control and an evaluation design.

7 Matching application

Matching application

Recruitment delayed-feedback funnel. A candidate moves through 4 stages: screening, interview, offer, outcome. Action at each stage is ADVANCE or REJECT. Reward is +1 if the hire materialises (only at the terminal stage); 0 otherwise.

Why MC? Episodes are short (4 transitions max), terminal, and the reward signal is fundamentally trajectory-level. There is no value in bootstrapping a TD target from a halfway-formed $\hat{V}$; just take the full return.

Off-policy evaluation in production. The current ATS routing policy logs (x_t, a_t, r_T) triples. To compare a candidate policy π without an A/B test, we use WIS over the logs. Two requirements:

1. Log propensities $b(a \mid x)$ at the time of action, not at analysis time. (Logging the action alone is not enough — if the production rules change between data collection and analysis you cannot reconstruct b.)
2. Inject a forced-exploration budget into the production policy so $b > 0$ on every reasonable (x, a). Coverage is the price of admission to off-policy evaluation.

synthetic insurance parallel. The same scaffold works for synthetic insurance's lead-broker matcher with the funnel reinterpreted as: contact attempt → quote produced → policy bound → renewed. Long horizons + delayed rewards give per-decision IS its single most useful application.

8 Exercises

See `workbook.ipynb`:

Ex. 1 ($\star$). First-visit returns from a sample trajectory.

Ex. 2 ($\star$). Every-visit MC prediction for a fixed Blackjack policy ("always stick").

Ex. 3 ($\star\star$). On-policy MC control with ϵ-soft policies; reproduce figure 5.2.

Ex. 4 ($\star\star$). Ordinary importance sampling.

Ex. 5 ($\star\star$). Weighted importance sampling.

Ex. 6 ($\star\star$). Empirical variance comparison OIS vs WIS.

Ex. 7 ($\star\star\star$, applied — matching). Off-policy MC on the recruitment funnel.

Ex. 8 ($\star\star$). Verify the bound $\mathrm{Var}(w \cdot r) \leq w_{\max}^2 \mathbb{E}[r^2]$.

9 Further reading

`extended_reading.pdf` unpacks Robbins–Monro (1951), Singh & Sutton (1996), Precup-Sutton-Singh (2000), Thomas-Brunskill (2016), and Joachims- Swaminathan-de Rijke (2018, BanditNet). Source material in `research/bandits_matching_sota.md` §3 and Sutton & Barto Chapter 5.